

- a) $\pi_{Baslik, yil} (KITAP)$
- b) $\sigma_{Bolum = "IIS"} (OGRENCI)$
- c) $\pi_{ogrID, kitapID} (OGRENCI \times KITAP)$
- d) $\sigma_{Yayinci = "SAÜ" \wedge yil < 1990} (KITAP)$
- e) $\pi_{YAdi} (\sigma_{Adres = "Sakarya"} (YAZAR))$
- f) $\pi_{ogrAdi} (\sigma_{Yas > 30 \wedge \neg (Bolum = "IIS")} (OGRENCI))$
- g) $\rho_{Adi} (YAZAR)$
- h) $\pi_{ogrID, ogrAdi, Bolum, Yas} (\sigma_{Bolum = "IIS"} (OGRENCI) \bowtie ODUNC)$
- i) $\pi_{Baslik} (\sigma_{YAdi = "Kotan"} (YAZILMIS) \bowtie KITAP)$

i)

$$\pi_{\text{Boslík}} \left(\left(\sigma_{\text{yAdi} = \text{"Koton"}} (\text{YAZILMIS}) \bowtie \text{KITAP} \right) - \left(\sigma_{\text{Anahar} = \text{"veritabanı"}} (\text{ACIKLAMA}) \bowtie \text{KITAP} \right) \right)$$

ii)

KITAP $\bowtie$ ACIKLAMA

k)

OGRENCI $\bowtie$ ODUNC $\bowtie$ KITAP

l)

$$\pi_{\text{Boslík}} \left(\sigma_{\text{yAdi} = \text{"YASIN"}} (\text{YAZILMIS}) \bowtie \text{KITAP} \right)$$

m)

$$\pi_{\text{yAdi}} \left(\sigma_{\text{ogrAdi} = \text{"Veli"}} (\text{OGRENCI}) \bowtie \text{ODUNC} \bowtie \text{YAZILMIS} \right)$$

n)

$$\pi_{\text{kitapID}} \left(\sigma_{\text{Anahar} = \text{"veritabanı"}} (\text{ACIKLAMA}) \right) \cap \pi_{\text{kitapID}} \left(\sigma_{\text{Anahar} = \text{"programlama"}} (\text{ACIKLAMA}) \right)$$